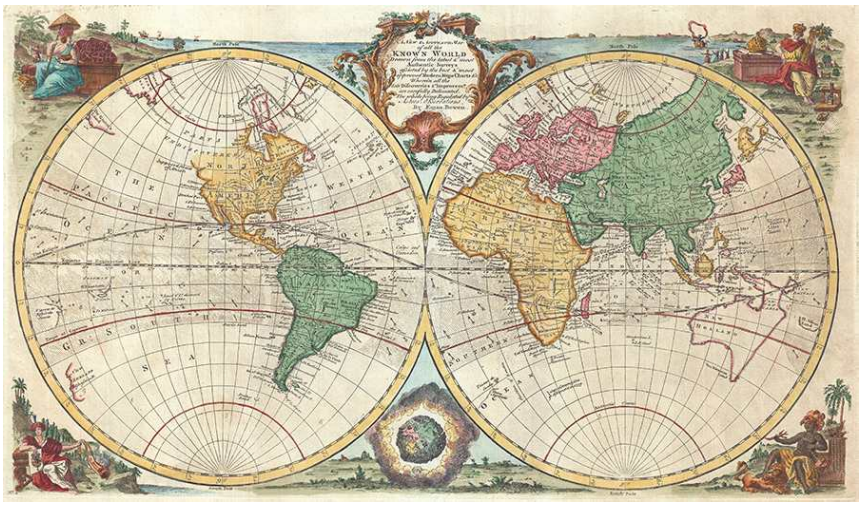


1744—Emanuel Bowen

Fig. 3.1. World maps from 1506 to 1744.

By 1924 it was crystal clear that the Newtonian laws also fail in the realm of the very small (molecules, atoms, and fundamental particles). To deal with this Niels Bohr, Werner Heisenberg, Erwin Schrödinger, and others gave us the *quantum laws of physics* (Figure 3.2). Using the concepts that everything fluctuates randomly at least a little bit (which I describe in Chapter 26), and that these fluctuations can produce new particles and radiation where before there were none, the quantum laws have brought us lasers, nuclear energy, light-emitting diodes, and a deep understanding of chemistry.